PROJECT DESIGN PHASE-II

Technology Stack

(Architecture & Stack)

DATE	TEAM ID	PROJECT NAME	MAXIMUM MARKS
29/09/2026	SWTID-2026-8965	Auto Ticket Classification using Flow Designer	4 Marks

Technical Architecture – Auto Ticket Classification using Flow Designer

This architecture shows how the different components work together to automatically classify IT support tickets using ServiceNow Flow Designer and send an email confirmation to the requester.

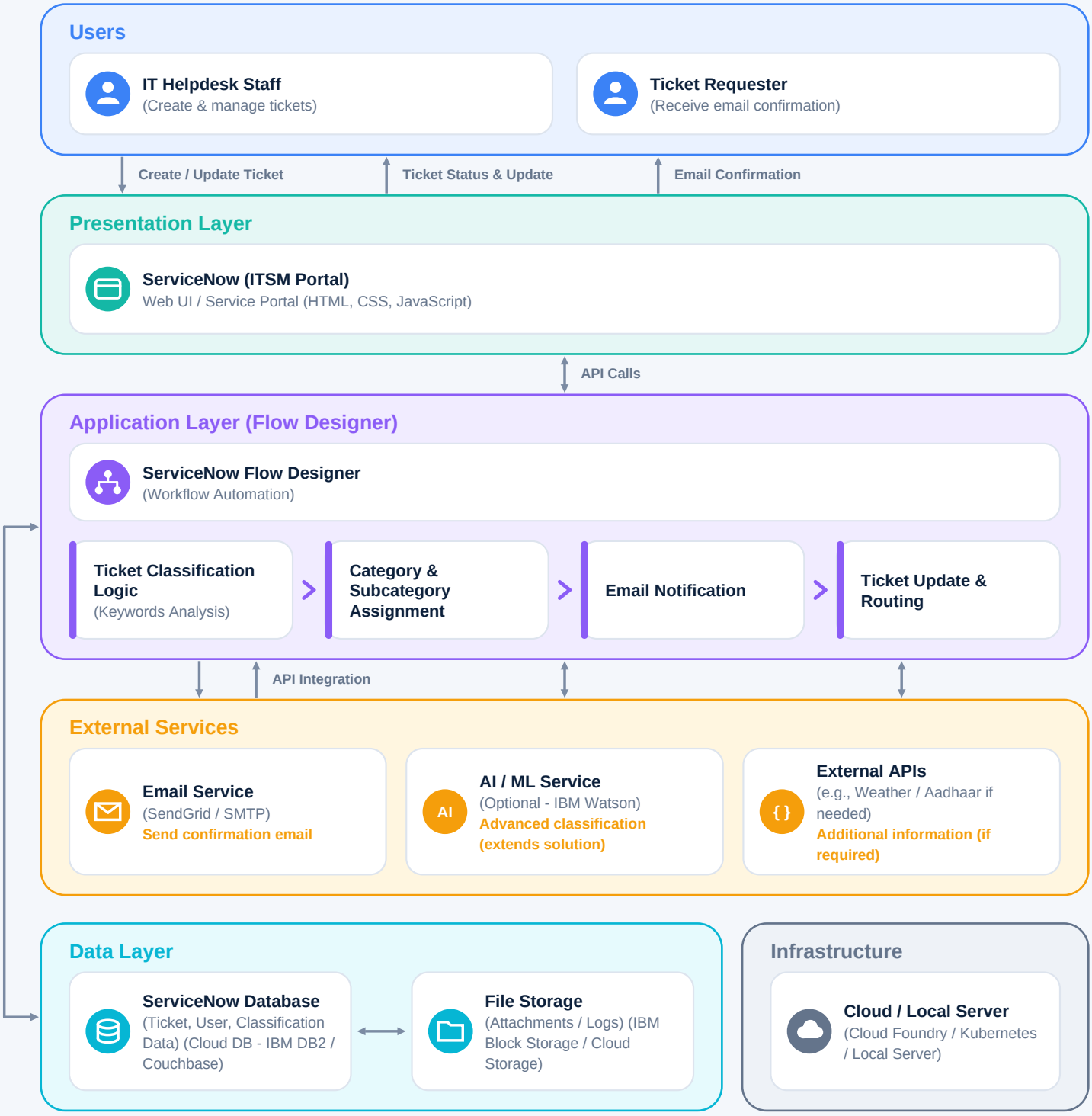

Table 1: Components & Technologies

Following are the components and technologies used in the proposed solution.

S.No	Component	Description	Technology
1	**User Interface**	Interface through which IT helpdesk staff create and manage tickets.	**ServiceNow Web UI / ServiceNow Portal**
2	**Application Logic-1**	Automatically triggers ticket classification when a new ticket is created.	**ServiceNow Flow Designer**
3	**Application Logic-2**	Analyzes keywords in the Short Description and determines the appropriate issue type.	**Flow Designer If / Else If Logic**
4	**Application Logic-3**	Updates Category and Subcategory and sends confirmation email.	**Flow Designer Update Record & Send Email Actions**
5	**Database**	Stores ticket records, caller information, category, subcategory, state and assignment details.	**ServiceNow Database / Tables**
6	**Cloud Database**	Database storage required by the ServiceNow platform.	**ServiceNow-managed Database**
7	**File Storage**	No separate file-storage component is required for the core ticket-classification workflow.	Not Required
8	**External API-1**	External API is not required for keyword-based ticket classification.	Not Used
9	**External API-2**	External API is not required for the proposed solution.	Not Used
10	**Machine Learning Model**	No machine-learning model is used; classification is performed using predefined keywords and Flow Designer conditions.	Not Used
11	**Infrastructure (Server / Cloud)**	Hosts the ServiceNow application, database and Flow Designer automation.	**ServiceNow Personal Developer Instance (PDI) / ServiceNow Cloud**

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	No separate open-source framework is required. The solution is developed using ServiceNow's built-in Flow Designer and platform configuration.	**ServiceNow Flow Designer**
2	**Security**	Ticket information should be accessible only to authorized users. The project includes validation of access control and security during testing.	**ServiceNow Access Controls / User Roles**
3	**Scalable Architecture**	The solution can be extended by adding new keywords, categories, subcategories and classification rules without redesigning the entire application.	**ServiceNow Flow Designer**
4	**Availability**	The application is hosted within the ServiceNow platform, allowing the ticket-classification workflow to be available through the ServiceNow instance.	**ServiceNow Cloud / PDI**
5	**Performance**	Classification is triggered when a ticket is created and uses predefined keyword conditions, reducing the need for manual ticket review.	**Flow Designer Automation**